SCCS 东南康州中文学校
Southeastern CT Chinese School

Introduction to Arduino Computing and Electronics

This laboratory-based course will introduce students to the world of Arduino computing. The Arduino is an inexpensive computer that excels in controlling electronic circuits. Arduino boards are able to read inputs (such as light on a sensor, temperature, a finger on a button or even a Twitter message) and turn it into an output (activating a motor or relay, turning on an LED, publishing an online message). The platform has been used extensively for monitoring sensors, robotics, home automation and other applications. Students will learn basic electronics, including how to construct a circuit on a breadboard, how to read a schematic diagram, the role of different electronic components in a circuit and how to design simple circuits. Students will also learn computer programming using the Arduino Integrated Development Environment, which uses a variant of the C computer language. Concepts to be covered include reading and constructing a computer flowchart, understanding and debugging code, learning how to use a compiler and principles of programming in C. There are no course prerequisites other than a genuine interest in science. Students must purchase the Elegoo Mega 2560 Project Starter Kit (EL-KIT-008 on Amazon - $60). Students must also have their own laptop computer running the Arduino IDE (available for download free at www.arduino.cc/en/Main/Software) with an available USB port. Due to the presence of numerous small parts in the kit, this course is only available for students in Grade 6 or above. Enrollment will be limited to 26 students.

Arduino Course Syllabus: SCCS 2020/2021

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Lesson 1: Introduction

- Introduction to hardware and how to connect it to your computer
- Introduction to Arduino IDE (Integrated Development Environment)
- Flowcharts and Sketches
- Basic syntax of a sketch

- Loading and running the Blink program. Varying parameters in Blink

Lesson 2: Variables

- What is a variable, and why would I want to use it?
- Different types of variables: Integers, Floating point variables, Character variables, String variables.
- Introduction to the Const (constant) modifier and why it is useful.
- Introduction to Unsigned and Long modifiers – managing memory.
- Appropriate use of variables to make code easier to read and modify.
- Modifying the Blink program to use variables and constants.
- Doing arithmetic operations with variables.

Lesson 3: Loops and Conditionals

- Introduction to For loops – simplifying with nested loops
- Using For to increment variables
- Manipulating variables arithmetically
- Difference between For loops and While loops.
- Conditionals: If, Then, Else – Interrupting a loop.

Lesson 4: Digital Outputs and Electronic Components (may need to divide first three lessons into four and introduce the concept of constants and #Define).

- Introduction to the breadboard – how it works
- Schematic diagrams – what they represent
- Basic LED circuit – voltage source, resistor, LED
- Resistors – types, schematic symbol, color code
- Diodes – types, schematic symbol, concept of polarization, focus on LEDs
- Wiring the basic LED circuit
- Ohm's Law
- Wiring the LED circuit to a digital output
- Blink with two LEDs

Lesson 5: Digital Inputs

- Introduction to digital inputs – why we use them
- Boolean variables
- Concept of pullup resistors
- Wiring a light switch with two buttons
- Variant that turns blinking LED on and off using pushbuttons

Exam 1 – On all materials through Lesson 5.

Lesson 6: Analog Inputs

- Why analog inputs are used

- Reading an analog input to a variable
- Variable resistors – introduction, schematic symbol
- Concept of voltage dividers and how they relate to variable resistors
- Experiment varying number of times an LED blinks based on position of a control (variable resistor).

Lesson 7: Analog Outputs

- Why one uses an analog output – controlling brightness of an LED as an example
- Concept of pulse width modulation – any output can be an analog output
- Using the variable resistor to change the brightness of an LED – Dimmer project
- Introduction to the RGB LED – How different colors can be formed from 3 LEDs
- RGB LED experiment – combining For loops to increment PWM outputs.

Lesson 8: Combined logic operations (and, or, not)

- Truth tables and simple logic operations
- Basic if, then, else operations using and, or and not logic.
- Demonstrating the truth table using two switches and an LED – how do AND, OR and NOT work?

Lesson 9: Libraries and the Serial Monitor

- Definition of an Arduino library and how to include them
- Introduction to the serial port – how to access it on the IDE, concept of baud rates, etc.
- Introduction to character variables
- Hello World program – using the serial port to receive messages
- Modifying code to use the serial monitor as a debugging output – Blink with serial monitor (no wiring example on pin 13).
- Receiving input from the serial port – character and string input variables
- Concept of Serial.read() and Serial.readString()
- Program to blink LED by the number of times received from the serial monitor

Exam: Midterm Exam – On all materials through Lesson 9

Lesson 10: The Switch/Case routine

- The Map function – turning continuous variables into discrete bins.
- Introducing Switch, Case – the computer's equivalent of a rotary selector switch.
- Demonstration of how Switch, Case can be more efficient than If, Then, Else if, Else if…
- Sensing temperature – introduction to the DHT-11 module.
- Using Switch and Case to build a thermostat.

Lesson 11: Subroutines – callable programs

- What is a subroutine and how do I call it?
- Demonstrating cleaner code with subroutines and custom functions.
- Introduction to photocells

- Making a light meter using photocells and custom functions.

Lesson 12: Transistors and Relays as switches – motor control with Arduino

- Introduction to relays: Electromechanical switchess
- Introduction to transistors – electronic switches. The chemistry of semiconductors and how a transistor works.
- Introduction to the Power Supply Module.
- Using a transistor to control the speed of a motor – via the serial port and/or via an attached variable resistor through an analog input.

Lesson 13: Introduction to capacitors – RC time constants and measuring time with Arduino

- Introduction to capacitors – devices for storing electrical charge.
- Why capacitors block DC and pass AC current.
- The RC time constant – building timers with passive components.
- Using comments to disable parts of programs – writing two programs in one.
- Introducing the serial plotter – graphing data.

Lesson 14: Introduction to integrated circuits – the Shift Register

- Definition of an integrated circuit
- How a shift register allows you to convert serial bits to parallel outputs
- Making an LED bargraph
- Definition of bits and bytes. Introduction to most significant and least significant bits
- Converting binary to decimal and vice versa.

Lesson 15: Beyond the delay function: Using Millis() to run multiple functions simultaneously.

- Introducing the Millis() function – a built in clock on every Arduino
- Revisiting Blink with the Millis() function – avoiding the Delay() command
- Multitasking using Arduino – running three blinking LEDs simultaneously using Millis() to time them.

Lesson 16: Classes and Object oriented programming

- Definition of objects and classes – an analogy to things we see in everyday life
- Introduction to member variables and member functions
- Introduction to object constructors
- Revisiting Blink using classes – Coding a class called Flasher
- Using two flasher objects to blink different LEDs at different rates

Exam: Exam 3 – On all materials through Lesson 16

Lesson 17: Arrayed variables

- Definition of Arrays
- Example of how we use arrays to encode and control a set of LEDs

- Using an LED matrix display
- Encoding and sending data using arrays of bytes

Lesson 18: More on Classes and Arrays – Multisegment LED displays and the membrane keypad

- How the membrane keypad works – an array of switches
- Understanding the keypad library
- Defining a keypad with character arrays
- Converting character variables to integers
- How a seven segment LED display works
- Using the seven segment LED display to count

Lesson 19: Characters and Strings

- The char variable type – characters and character arrays
- Different methods to define a string of characters
- Multidimensional character arrays
- The sprintf() function – building strings from pieces
- Making a calendar using character arrays and strings

Lesson 20: Comparing Strings

- String objects: A simpler way to manipulate character strings
- String functions – manipulating and comparing strings
- Reading strings from the serial monitor
- Converting variables to strings. Converting numerical bases (binary, decimal, hexadecimal) using strings.

Lesson 21: Multisegment Displays

- How the multisegment display works
- Declaring seven segment displays as objects
- Using hexadecimal byte arrays to define digital numbers

Exam: Final Exam